

Vermikendra

Product and Technical Requirements

A solar-powered, off-grid monitoring system that keeps village compost worms alive, tells operators when compost is ready, and records every kilogram of waste turned into fertilizer.

DOCUMENT	TITLE	VERSION	DATE	PREPARED BY
VK-PRD-001	Product Requirements	0.1 draft	14 Sep 2026	Team Vermikendra, Rashtriya Raksha University
VK-TRD-001	Technical Requirements	0.1 draft	14 Sep 2026	Team Vermikendra, Rashtriya Raksha University

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VK-PRD-001

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Product Requirements Document

DOCUMENT ID	VK-PRD-001
VERSION	0.1, draft for team review
DATE	14 September 2026
PREPARED BY	Team Vermikendra, Rashtriya Raksha University
EVENT	Hack for Social Cause, Youth Tech Challenge 2027 (VBYLD 2027)
PRIMARY THEME	Rural Development & Livelihoods
SECONDARY THEMES	Environment & Sustainability; AI, GIS & Emerging Technologies
COMPANION DOCUMENT	VK-TRD-001, Technical Requirements
SUBMISSION DEADLINE	15 October 2026 (team target: 14 October)

1. Summary

Vermikendra is a solar-powered, off-grid monitoring system for village vermicompost units. Sensors in each compost bin track bed temperature at five points, moisture, microbial respiration (CO2) and weight. A long-range LoRa radio sends the data to a small gateway at the panchayat or self-help group (SHG) office. The gateway shows a simple dashboard on any phone over local WiFi, with no internet and no SIM card needed.

It does three jobs:

1. **Keeps worms alive** by warning before heat, dryness or foul air kills them.
2. **Tells the operator when compost is ready**, using the respiration trend instead of guesswork.
3. **Records how much waste went in and how much compost came out**, in kilograms, automatically.

Vermikendra keeps village compost worms alive and turns crop and dairy waste into measured, sellable fertilizer, with no internet needed.

2. Problem

2.1 Context

- After harvest, crop residue is often burned in the field because it is the fastest way to clear land for the next crop. Dairy dung and kitchen waste are often dumped.
- Vermicomposting, where earthworms convert organic waste into fertilizer, is a well-known, low-cost alternative that produces sellable products (vermicompost and vermiwash).
- Government programmes already promote organic waste management and organic manure, and many compost units are run by women's SHGs.

2.2 Why village units fail

Compost worms (commonly *Eisenia fetida*) survive only in a narrow band of conditions.

FAILURE	WHAT HAPPENS	WHY NOBODY NOTICES
Heat	Fresh dung and piled straw heat up while decomposing. Worms die or flee once the bed passes roughly 35°C.	The surface can feel fine while a lower layer overheats.
Dryness or waterlogging	Worms need a moist bed, roughly 60 to 80% moisture. Too dry and they die; too wet and air is cut off.	Changes happen over one hot afternoon or one rainy night.
No air (anaerobic bed)	The bed turns sour, releases ammonia and rotten-egg smells, and worms die.	Noticed only once the smell is strong.
Harvest guesswork	Compost is emptied too early (poor quality) or left too long (lost capacity).	There is no objective readiness signal outside a laboratory.
No records	No proof of how much waste was processed or compost produced.	Buyers, lenders and scheme officers cannot verify output.

These thresholds are initial values from vermicomposting practice and will be tuned with pilot data (TRD section 7).

2.3 Problem statement

Village vermicompost units fail silently: operators cannot see heat, moisture and air problems inside the bed until worms are already dying, cannot tell when compost is ready, and cannot prove their output.

2.4 Claims we will and will not make

WE WILL SAY	WE WILL NOT SAY
Vermikendra reduces unit failure and gives crop and dairy waste a reliable, paid alternative to burning or dumping.	"Vermikendra solves stubble burning."
Readiness is estimated from the CO2 respiration trend, the same principle used in compost maturity tests.	"We measure microbial activity directly" or "we certify compost quality."
Weighed records support reporting, sales and scheme claims.	"Our records replace laboratory testing under the Fertiliser Control Order."
Figures come from our own pilot data or cited sources.	Any statistic we have not sourced.

3. Goals and non-goals

3.1 Product goals

- **G1** Prevent worm mass-mortality events through early, actionable alerts.
- **G2** Give a readiness estimate an operator can act on.
- **G3** Produce automatic, weighed input and output records per bin and per batch.
- **G4** Work fully offline and off-grid, in the operator's language.
- **G5** Be cheap to replicate: one gateway serves a cluster of bins.

3.2 Hackathon goals

- **H1** Submit a complete entry by 14 October 2026: deck, video and public repository.
- **H2** Place in the top 3 for Gujarat in the state round (26 October to 30 November 2026).

- **H3** Be selected as a national finalist (1 to 15 December 2026).
- **H4** Place in the national top 5 at VBYLD 2027 (10 to 12 January 2027).

3.3 Non-goals

- Soil pH, NPK or heavy-metal sensing, and laboratory certification of compost.
- Logistics of collecting residue from fields (balers, transport).
- A marketplace or payments.
- Cloud-only operation. The cloud is optional sync, never a dependency.
- Acoustic sensing of worm activity with the INMP441 microphone (a research idea, not in scope).

4. Users

PERSONA	WHO	CONTEXT	NEEDS	PAIN TODAY
Operator (primary)	SHG member or farmer running 2 to 10 bins	Shaded shed near the village; basic Android phone; Gujarati or Hindi; unreliable power and internet	Know which bin needs attention and what to do; know when to harvest	Worm losses, guesswork, no records
Supervisor	SHG cluster coordinator, panchayat secretary, block-level scheme staff	Oversees several units and reports upward	Proof that units work; output figures for reports	Paper registers; site visits to verify
Buyer	Farmer, farmer producer organisation (FPO), nursery	Buys compost in bulk	Consistent quality; traceable batches	Cannot tell a good batch from a poor one
Extension expert	Krishi Vigyan Kendra (KVK) or agriculture university staff	Trains operators	Data to tune practice; remote view of problem units	Problems reported late

Design implications:

- Status must be readable at a glance by a first-time, low-literacy user: colour plus icon plus a word, never colour alone.
- Every alert carries one plain action.
- The interface works in Gujarati, Hindi and English.

5. Alignment

5.1 Hackathon themes

THEME	HOW VERMIKENDRA FITS
Rural Development & Livelihoods (primary)	Agri-tech for SHG-run compost enterprises; women entrepreneurs; higher, more reliable output
Environment & Sustainability	Circular economy for crop, dairy and kitchen waste; less open burning and dumping
AI, GIS & Emerging Technologies	Readiness prediction, heat-risk forecasting, LoRa IoT; site mapping (P2)

5.2 Government programmes

Scheme details change. Verify current guidelines before quoting any figure in the pitch.

PROGRAMME	LINK TO VERMIKENDRA
Swachh Bharat Mission (Grameen) Phase II	Biodegradable waste management in villages; weighed records show waste processed
GOBARdhan	Cattle dung and organic waste to wealth; dung is a core feedstock
DAY-NRLM and Lakhpati Didi	SHG livelihood enterprises; income from compost sales
PM-PRANAM	Reducing chemical fertilizer use through organic alternatives
National Mission on Natural Farming	Demand for on-farm organic inputs
Crop residue management efforts	A productive destination for baled residue instead of burning
Mission LiFE	Community-level sustainable practice

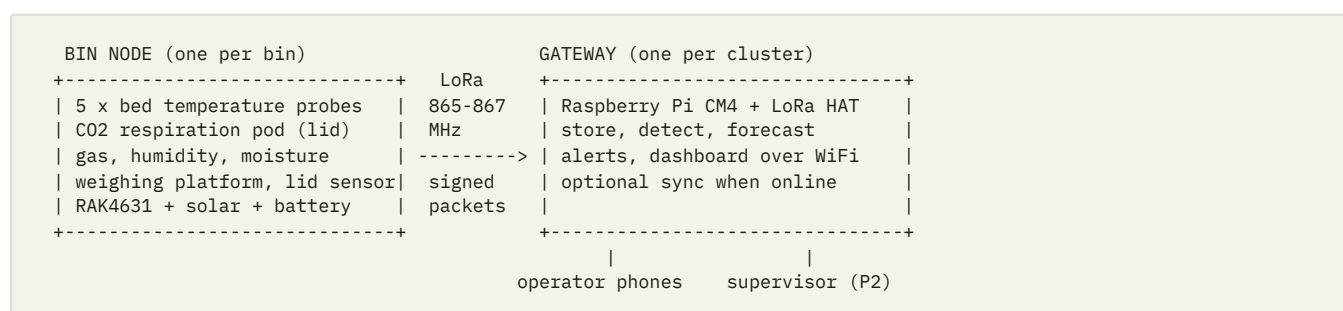
5.3 Judging criteria

CRITERION	OUR EVIDENCE
Relevance	Direct fit to three themes and to existing programmes
Technical Strength	Multi-sensor LoRa node, respiration measurement cycle, edge failsafe, offline gateway, prediction model
Functionality	A real bin with live worms, multiple weeks of real data, live alerts during the demo
Creativity	CO2 respiration as a readiness signal; lid-plus-weight automatic records; one gateway for a cluster
Social Cause Impact	Livelihoods, kilograms of waste diverted, worm losses prevented
Presentation	Before-and-after story, live dashboard, interface in the operator's language
Teamwork	Three clear workstreams: hardware and firmware, gateway and data, product and field

6. Solution

6.1 How it works

- 1. Sense.** A solar-powered node on each bin reads bed temperature at five points, moisture, gas, humidity, weight and lid movement every 5 minutes, and runs a CO2 respiration cycle every 4 hours.
- 2. Send.** The node transmits a small signed packet over LoRa in the 865 to 867 MHz band (license-exempt in India) to the gateway, up to a few kilometres away.
- 3. Decide.** The gateway stores readings, detects feeding and harvest events, forecasts heat risk, estimates readiness and raises alerts. The node also runs basic safety rules on its own when the gateway is unreachable.
- 4. Show.** Operators connect their phone to the gateway's WiFi hotspot and open the dashboard. Alerts also go out by Telegram when internet is available (P1).



6.2 What is new

- 1. **Depth-resolved heat warning.** Five probes catch hot layers that a single sensor misses, with a forecast of when a layer will cross the danger line.
- 2. **Respiration-based readiness.** A short closed-lid CO2 measurement gives a respiration rate. Its decline and plateau indicate maturity.
- 3. **Automatic weighed records.** A lid opening plus a weight step becomes a logged feeding (kg in) or harvest (kg out) without manual entry.
- 4. **Off-grid cluster network.** One gateway serves many bins without WiFi, SIM cards or cloud.
- 5. **Edge failsafe.** The node protects the bed with local rules even if the gateway is down.

6.3 Alternatives

APPROACH	STRENGTHS	GAPS
Manual thumb rules and training	Free, familiar	Misses hidden hot layers and night events; no records
Generic "smart compost bin" (one temperature and humidity sensor, WiFi)	Cheap	One point only; needs WiFi; no readiness or weight
Laboratory maturity and nutrient testing	Authoritative	Slow, costly, one-off; no early warning
Industrial compost probes	Robust	Built for large windrows; expensive; usually temperature only
Vermikendra	Multi-parameter, predictive, automatic records, offline, cluster-scale	Needs pilot data to tune thresholds; not a laboratory certificate

7. User journeys

J1 Morning check (operator). She opens the dashboard on her phone over the gateway hotspot and sees four bins: three green, one amber. She taps the amber bin: "Bed is drying (58%). Sprinkle water evenly and check again in an hour." She marks the action done.

J2 Afternoon heat (operator, critical case). In May, probe 4 in bin 3 rises from 29°C to 31°C in an hour. The node switches to 1-minute sampling. The gateway forecasts that the layer will cross 33°C in about an hour and raises a critical alert: a dashboard banner, the gateway buzzer, and a Telegram message if online (P1). Bed moisture is 70%, so the gateway triggers a short misting cycle (P1). If the bed were already wet, the alert would instead say "Shade the bin and do not add fresh dung today." The recovery is logged.

J3 Feeding (operator). She lifts the lid, adds kitchen waste and aged dung, and closes the lid. Within 90 seconds the bin timeline shows "Input +6.2 kg". She can tag the feedstock type.

J4 Harvest (operator). The bin shows "Maturing", then "Ready". She harvests; the weight drop is logged as "Output 18.4 kg" and the batch record closes.

J5 Monthly report (supervisor). He opens the Records page: waste processed, compost produced, alerts raised and resolved, node uptime. He exports a CSV for the block report.

8. Requirements

Priority: **P0** is required for the 14 October submission. **P1** is required by the state round (from 26 October). **P2** is targeted for the national showcase (January 2027).

8.1 Functional requirements

ID	REQUIREMENT	PRIORITY	ACCEPTANCE CRITERIA
FR-01	Measure bed temperature at 5 points	P0	Every 5 min, and every 1 min while any probe reads 30°C or more; within $\pm 0.5^{\circ}\text{C}$ of the TMP117 reference after calibration; faulty or disconnected probes flagged
FR-02	Heat and cold alerts with a recommended action	P0	Critical heat alert visible within 2 min of the condition; every alert shows bin, probe, value and action
FR-03	Measure bed moisture with dry and wet alerts	P0	Reading every 5 min; within ± 10 percentage points of a gravimetric check after calibration
FR-04	Run a CO2 respiration cycle and chart the trend	P0	Cycle every 4 h (configurable); rate reported with a quality flag; fresh feedstock shows a clearly higher rate than mature compost in a bench test
FR-05	Weigh the bin and detect feeding and harvest events	P0	Mass reported every 5 min; a 2.0 kg addition with the lid opened is logged as an input within ± 0.2 kg
FR-06	Show a readiness status (rules-based)	P0	Status of Not ready, Maturing or Ready, with the conditions met shown
FR-07	Serve the dashboard offline from the gateway	P0	A phone joins the gateway WiFi and loads the dashboard with no internet
FR-08	Bin overview with at-a-glance status	P0	Each bin shows colour, icon and word status, last update time and top alert
FR-09	History charts per bin	P0	Temperature (all probes), moisture, respiration, mass; 24 h, 7 d and 30 d ranges
FR-10	CSV export	P0	Readings, events and alerts for a chosen bin and date range
FR-11	Off-grid node power	P0	Node runs from solar and battery alone (see NFR-01)
FR-12	Replay mode for demos and reviewers	P0	The gateway replays a recorded dataset so the dashboard runs without hardware
FR-13	Node health view	P0	Battery voltage, signal strength, last seen, sensor faults
FR-14	Manual event logging fallback	P0	Operator can log input, harvest or turning by hand
FR-15	Language support	P0 status and alert text; P1 full interface	Gujarati, Hindi and English, switchable per user
FR-16	Automatic misting with manual override and lockout	P1	Pump runs only within configured limits; operator can stop or trigger it; every run logged
FR-17	Foul air (anaerobic) risk indicator	P1	Flags a sustained gas-resistance drop at high humidity, with action text
FR-18	Days-to-ready prediction	P1	Shows an estimate with a range once enough respiration data exists; hidden otherwise
FR-19	Store-and-forward on the node	P1	No data loss for gateway outages up to 24 h
FR-20	Remote alerts when internet is available	P1	Telegram message within 2 min of a critical alert; queued while offline
FR-21	Batch records and impact totals	P1	Per batch: start, inputs, days, output, readiness history; totals per site
FR-22	Spoken alerts in the local language at the gateway	P2	Critical alerts announced through a speaker

ID	REQUIREMENT	PRIORITY	ACCEPTANCE CRITERIA
FR-23	QR batch record for buyers	P2	A QR code on the compost bag opens a read-only batch summary
FR-24	Multi-site supervisor view with cloud sync	P2	Gateways sync when online; a supervisor sees all sites
FR-25	Low-cost satellite node design	P2	Temperature and moisture node design with a costed bill of materials

8.2 Non-functional requirements

ID	REQUIREMENT	TARGET
NFR-01	Node autonomy with no sun	At least 7 days (the design estimate is well above this; confirmed in testing)
NFR-02	Radio range, node to gateway	At least 1 km verified in the field; target 3 km in open rural terrain
NFR-03	Data completeness	At least 95% of scheduled readings (P0); at least 99% with store-and-forward (P1)
NFR-04	Alert latency	Critical alert visible within 2 min of the condition
NFR-05	Offline operation	All P0 features work with zero internet
NFR-06	Environment	Node electronics in an IP65-class enclosure; operation in 5 to 45°C shade temperature; sensors tolerate high humidity and ammonia
NFR-07	Usability	A first-time user finds the bin that needs attention, and its action, within 10 seconds
NFR-08	Accessibility	Status never shown by colour alone; touch targets at least 44 px; readable at arm's length
NFR-09	Security	Packets authenticated per node with replay protection; misting and settings need a PIN
NFR-10	Privacy	Store only operator name, role, language and phone number; minimise personal data in line with the Digital Personal Data Protection Act, 2023
NFR-11	Openness	Public repository under an OSI-approved licence; setup reproducible from the README
NFR-12	Cost	Prototype built mainly from owned parts; satellite node target of ₹3,000 or less (estimate, to be validated)
NFR-13	Maintainability	Sensors replaceable through connectors without soldering; configuration changes without reflashing

9. Success metrics

9.1 Pilot metrics (campus pilot from late September 2026)

METRIC	TARGET	HOW MEASURED
Worm mass-mortality events in the monitored bin	Zero over 30 days	Weekly worm check by the operator
Alert precision	At least 80% of warning and critical alerts confirmed real	Operator confirms or dismisses each alert
Heat warning lead time	At least 1 h before a probe reaches 33°C	Alert log compared with probe data
Data completeness	At least 95%	Received vs scheduled readings
Event capture	At least 90% of feedings logged automatically	Manual register compared with detected events

METRIC	TARGET	HOW MEASURED
Readiness estimate error (P1)	Within ± 5 days	Predicted date vs the actual harvest decision
Operator task time	Under 10 s to find the bin that needs attention	Timed test with 3 people outside the team

9.2 Hackathon milestones

MILESTONE	DATE
All members registered on MY Bharat; eligibility confirmed	20 Sep 2026
Live data flowing from a real bin	27 Sep 2026
Submission complete (team target)	14 Oct 2026
Official submission deadline	15 Oct 2026
State round	26 Oct to 30 Nov 2026
National finalist selection	1 to 15 Dec 2026
Mentoring by IIT Bombay	16 Dec 2026 to 9 Jan 2027
National showcase, New Delhi	10 to 12 Jan 2027

10. Release plan

RELEASE	DATE	SCOPE
R0, submission	14 Oct 2026	All P0 requirements; at least two weeks of real bin data; deck, video, repository
R1, state round	From 26 Oct 2026	P1: misting, foul air indicator, days-to-ready, store-and-forward, Telegram, batches, full Gujarati interface
R2, national	Jan 2027	P2: spoken alerts, QR batch record, multi-site view, satellite node design; field pilot with an SHG or KVK

The week-by-week build plan with owners is in TRD section 12.

11. Assumptions, dependencies and risks

Assumptions

- Rashtriya Raksha University meets the "AICTE-registered institution" eligibility rule. To be confirmed with the MY Bharat helpdesk.
- A campus site with shade, water and a steady supply of food or garden waste is available for the pilot bins.
- Healthy compost worms and aged dung can be sourced locally within a week.

Dependencies

- A CM4 carrier board with a 40-pin header. It is not on the purchase invoices.
- The parts listed in TRD section 3.2.
- Radio compatibility between the RAK4631 and the Waveshare LoRa HAT (TRD risk R2).

Top risks (full register in TRD section 11)

RISK	IMPACT	MITIGATION
Gateway board or radio incompatibility delays integration	Blocks the demo	Spike tests in the first three days; portable gateway software; fallback radio
Worms die before we collect data	No real dataset	Two bins, aged feedstock, shade; a failure the system detects is itself evidence
Not enough data for prediction by 14 Oct	Weaker AI claim	Rules-based readiness in R0; prediction in R1; clearly labelled
A public repository counts as public disclosure	Could block a later patent	Decide on a provisional patent before the repository goes public
Judges question the cost per bin	Credibility	Reference bin plus low-cost satellite nodes, with a costed design

12. Open questions

QUESTION	OWNER	NEEDED BY
Does RRU count as AICTE-registered for this track?	Team lead	18 Sep
Which WisBlock base board revision do we have (RAK19007 or older)?	Hardware	16 Sep
Is a CM4 carrier board available in a campus lab?	Hardware	16 Sep
Do we file a provisional patent before publishing the repository?	Team lead, with the university IPR cell	30 Sep
Which campus waste stream (mess or garden) feeds the pilot, and who tends the bins daily?	Product and field	18 Sep
Apache-2.0 or MIT licence? Depends on the patent decision.	Team lead	30 Sep
Which SHG, KVK or panchayat partners for the R2 field pilot?	Product and field	15 Nov

13. Submission checklist

- ☐ Presentation deck: 6 to 7 slides, 10 MB or less
- ☐ Demo video: 3 to 5 minutes, at least 720p, 80 MB or less, covering team, problem, prototype, user interaction and impact
- ☐ Public GitHub repository: complete source, README with setup instructions, architecture documentation, open-source licence
- ☐ Team details, self-certification and ID proof for every member
- ☐ Every member registered individually on the MY Bharat portal
- ☐ All declarations and disclosures required by the official guidelines
- ☐ Replay dataset in the repository so reviewers can run the dashboard without hardware

Glossary

TERM	MEANING
Vermicompost	Fertilizer produced when earthworms digest organic waste

TERM	MEANING
Vermiwash	Liquid drained from a vermicompost bed, used as a plant spray
<i>Eisenia fetida</i>	Red wiggler, the most common composting worm
Respiration rate	How fast microbes in the bed release CO ₂ : high in fresh waste, low in mature compost
LoRa	Long-range, low-power radio technology
Gateway	The central unit that receives node data and hosts the dashboard
Headspace	The air gap between the compost bed and the lid
SHG	Self-help group
KVK	Krishi Vigyan Kendra, a district agricultural extension centre
FPO	Farmer producer organisation

References

- Hack for Social Cause 2027 rules and themes: mybharat.gov.in/pages/hack_social
- 2026 national winners list: [National-Winner-HSC.pdf](#)

Technical Requirements Document

DOCUMENT ID	VK-TRD-001
VERSION	0.1, draft for team review
DATE	14 September 2026
PREPARED BY	Team Vermikendra, Rashtriya Raksha University
COMPANION DOCUMENT	VK-PRD-001, Product Requirements
STATUS	Items marked VERIFY must be checked against the physical part or its datasheet before wiring

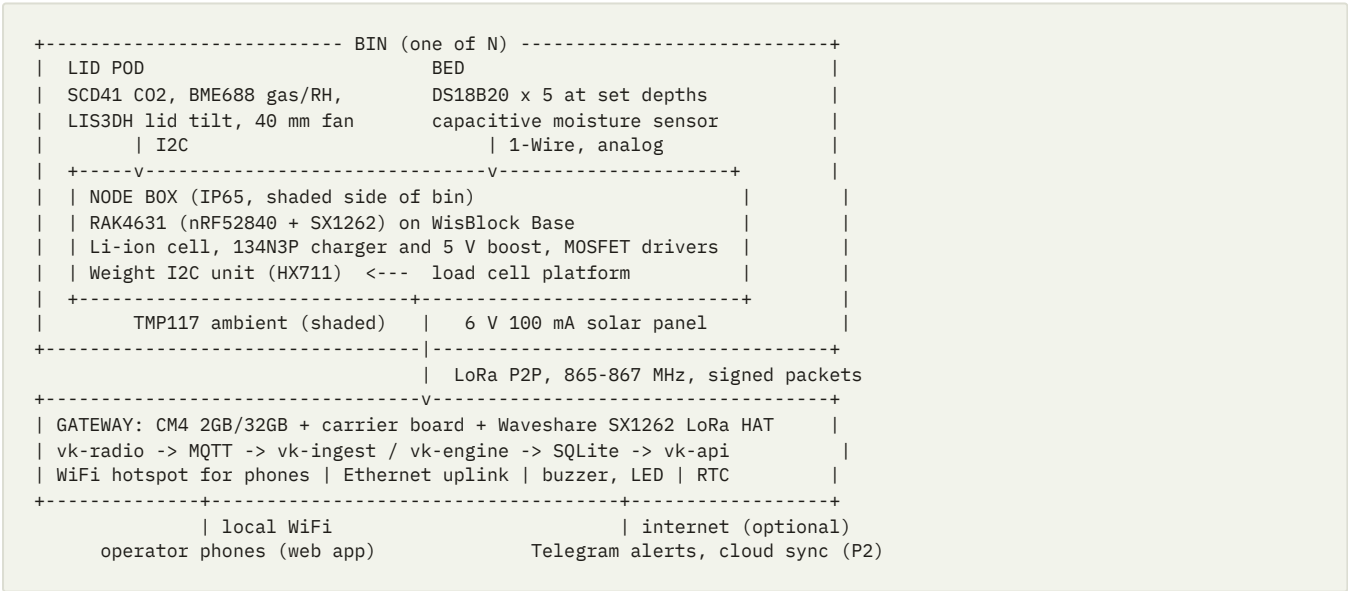
1. Purpose and scope

This document specifies how Vermikendra is built: hardware, radio link, node firmware, gateway software, algorithms, calibration, testing and the build plan. Requirement IDs (FR, NFR) refer to VK-PRD-001.

Units: temperature in °C, mass in kg (grams inside packets), CO2 in ppm, moisture as gravimetric moisture content (wet basis) after calibration, battery in volts.

2. Architecture

2.1 System context



2.2 Components

COMPONENT	RESPONSIBILITY	RUNS ON
Bin node	Sample sensors, run respiration cycle, detect lid events, apply edge failsafe, transmit	RAK4631 on WisBlock Base
Lid pod	Hold CO2, gas and lid sensors in the headspace; flush fan	Small IP65 box under the lid

COMPONENT	RESPONSIBILITY	RUNS ON
Weighing platform	Measure total bin mass	Four load cells between two boards
Gateway	Receive, store, analyse, alert, serve dashboard	Raspberry Pi CM4
Dashboard	Show status, actions, charts, records	Operator's phone browser (installable web app)
Cloud sync (P2)	Multi-site supervisor view	Any hosted service

2.3 Data flow

1. The node wakes on its timer, or on a LIS3DH interrupt when the lid moves.
2. It switches on sensor power (3V3_S), reads all sensors, and switches sensor power off.
3. It builds the payload, signs it, and transmits.
4. It opens a 2-second receive window for the acknowledgement and any command.
5. The gateway's radio service checks the signature and sequence number, decodes the packet and publishes JSON to MQTT.
6. The ingest service stores the reading. The engine computes calibrated values, events, respiration rate, heat forecast and readiness, and evaluates alert rules.
7. The API pushes updates to open dashboards over WebSocket. The notifier sounds the buzzer and sends Telegram messages when online.
8. Commands (misting, configuration) travel back through MQTT to the radio service and ride on the next acknowledgement.

2.4 Design principles

- **Offline first.** Every P0 feature works with no internet.
- **Safety at the edge.** The node protects the bed on its own if the gateway is down.
- **Portable gateway.** The gateway software runs on any Linux machine with the LoRa radio attached over USB, so a laptop can stand in for the CM4.
- **Trends over absolute claims.** Respiration and gas readings are used as trends against the bin's own history, not as laboratory measurements.
- **Replicable.** Commodity parts, open source, documented calibration.

3. Hardware

3.1 Owned inventory

From Robu invoice INV2627/203030 and two Amazon orders, all dated 20 August 2026.

PART	QTY	COST (₹, INCL. GST)	ROLE
RAKwireless RAK4631 WisBlock core (nRF52840 + SX1262), IN865	1	2,999	Node microcontroller and LoRa radio
RAKwireless WisBlock Base	1	1,639	Node carrier: battery and solar inputs, switched 3V3_S rail
Waveshare SX1262 868M LoRa HAT (UART, on-board CP2102 USB)	1	2,809	Gateway radio

PART	QTY	COST (₹, INCL. GST)	ROLE
Raspberry Pi CM4 (CM4102032: 2 GB RAM, 32 GB eMMC, wireless)	1	11,275	Gateway computer
CM4 heatsink	1	299	Gateway cooling
DFRobot Gravity SCD41 CO2 sensor (400 to 5,000 ppm)	1	6,539	Respiration measurement
SmartElex BME688 environmental sensor	1	1,019	Gas resistance, humidity, pressure in the lid pod
SmartElex TMP117 high-precision temperature sensor	1	162	Ambient reference and probe calibration reference
DS18B20 waterproof temperature probe, 1 m	5	455	Bed temperature at five points
Adafruit LIS3DH accelerometer	1	799	Lid open and close detection; wake interrupt
M5Stack Weight I2C Unit (STM32 + HX711)	1	609	Load cell interface
INMP441 I2S MEMS microphone	1	149	Not used in the MVP
Waveshare industrial USB hub (switchable dual host)	1	2,139	Gateway peripherals: radio over USB, backup storage
865 to 868 MHz rubber duck antenna, 1.8 dBi	1	125	Node antenna
IPEX to RP-SMA female pigtail, 20 cm	1	107	Node antenna bulkhead
PG7 polyamide cable gland	4	40	Enclosure cable entries
Lever splice connector (PCT-SPL-42)	2	82	Field wiring joints
Dupont jumper wires, female to female, 40	1	41	Bench wiring
CAT6 patch cable	1	158	Gateway uplink
JST-PH 2.0 mm 4-pin cable set, pack of 2	10	890	Gravity I2C sensor leads
6 V 100 mA solar panel with 5 V boost and Li-ion charger board (134N3P)	1	99	Node power

Totals: Robu ₹32,251.20 including ₹806.20 shipping; Amazon ₹1,125.00 including shipping and fees.

3.2 Parts to acquire

Prices are rough market estimates. Check before ordering.

PART	QTY	EST. ₹	PRIORITY	WHY
CM4 carrier board with 40-pin header and RTC (for example Waveshare CM4-IO-BASE or the official CM4 IO Board)	1	2,000 to 5,000	P0 , blocker	The CM4 cannot boot or host the HAT without one. Skip if a lab can lend one.
5 V 3 A power supply for the carrier	1	400 to 600	P0	Gateway power
18650 Li-ion cell (2,600 mAh or more) with holder and JST-PH 2.0 lead	2	300 to 600	P0	Node battery and a spare
50 kg half-bridge load cells with combinator board	4 + 1	300 to 500	P0	Weighing platform
Capacitive soil moisture sensor (v1.2 or v2.0, 3.3 V compatible)	2	150 to 300	P0	Bed moisture, plus a spare

PART	QTY	EST. ₹	PRIORITY	WHY
40 mm 5 V DC fan	1	100 to 150	P0	Headspace flush
Logic-level N-MOSFET modules, or AO3400 or IRLZ44N with flyback diodes	3	100 to 200	P0	Switch fan, pump and 5 V loads
IP65 ABS junction boxes, about 150 x 110 x 70 mm and 100 x 68 x 50 mm	2	300 to 500	P0	Node box and lid pod
Breathable ePTFE vent (adhesive patch or M12 vent plug)	2	100 to 300	P0	Keeps liquid water off the pod sensors
PG7 cable glands (extra)	8	80	P0	Four are not enough (section 3.6)
4.7 kΩ resistors, perfboard, heat-shrink, silicone sealant, desiccant, cable ties	1 lot	150 to 250	P0	1-Wire pull-up and assembly
Buzzer and LED for the gateway	1	30	P0	Local alert
DS3231 RTC module, only if the carrier has no RTC	1	150	P1	Offline timekeeping
Mini water pump or solenoid, tubing, misting nozzle, separate supply	1 set	300 to 700	P1	Automatic misting
Spare BME688	1	1,000	P1	Harsh environment
Opaque compost bins, 40 to 60 L, with drainage holes and tray	2	500 to 800	P0	One monitored bin, one hand-checked reference bin
Compost worms (<i>Eisenia fetida</i>), about 0.5 kg per bin	1 kg	varies	P0	Source from a local KVK or compost unit
Shade net, dry bedding (leaves, shredded cardboard), aged cow dung	1 lot	low	P0	Bed preparation

Estimated total: ₹6,000 to 11,000, depending mostly on whether a carrier board can be borrowed.

3.3 Node electrical design

3.3.1 POWER AND SWITCHING

```

6 V panel --> 134N3P charger --> 1S Li-ion cell --> Base PHR-2 input --> LDO 3.3 V --> RAK4631
      |                                     |
      |                                     +--> 3V3_S (WB_IO2) --> I2C sensors,
      |                                     DS18B20, moisture
      +--> 5 V boost --+--> MOSFET (WB_IO4) --> 40 mm fan
                        +--> load switch (WB_IO6) --> Weight I2C unit [VERIFY supply]

```

- The base board's own charger stays idle because nothing is connected to its solar or USB input. The 134N3P board charges the cell from the panel.
- The base board's solar connector expects a 5 V panel. Do not connect the 6 V panel to it directly.

3.3.2 I2C BUS

DEVICE	ADDRESS	SUPPLY	NOTES
SCD41 (DFRobot Gravity)	0x62	3.3 to 5 V	Lid pod
BME688	0x76 or 0x77	3.3 V	Lid pod; VERIFY the breakout's address setting
LIS3DH	0x18 (0x19 alternative)	3 to 5 V	Lid pod; INT1 to WB_IO3

DEVICE	ADDRESS	SUPPLY	NOTES
TMP117	0x48	1.8 to 5.5 V	Ambient, in a shaded ventilated spot
M5Stack Weight I2C Unit	0x26	5 V through its HY2.0-4P port, VERIFY	Rated 0 to 40°C. VERIFY its I2C pull-ups go to 3.3 V, not 5 V, before connecting to the nRF52840

Run the bus at 100 kHz and keep total cable length under about 1 m. Each breakout adds its own pull-ups. If the combined pull-up resistance falls below about 1.5 kΩ, remove some.

3.3.3 PIN MAP

Logical WisBlock names. Not every WB pin is broken out on every base board. **VERIFY** against the base board pinout; a WisBlock IO adapter module exposes the rest if needed.

SIGNAL	WISBLOCK PIN	NOTES
I2C SDA, SCL	I2C1	All I2C devices
Sensor power	3V3_S, enabled by WB_IO2	High = on; off during sleep
1-Wire bus (5 x DS18B20)	WB_IO1	4.7 kΩ pull-up to 3V3_S; powered (3-wire) mode, not parasite mode
LIS3DH INT1	WB_IO3	Wake on lid tilt
Fan MOSFET gate	WB_IO4	100 Ω series resistor, 100 kΩ pull-down
Pump MOSFET gate (P1)	WB_IO5	Same as fan; flyback diode across the pump
5 V load switch (weight unit)	WB_IO6	Power-gates the weight unit
Moisture sensor	WB_A1	Analog input
Battery voltage	WB_A0	On-board divider used by RAK4631 examples; VERIFY for this base board

3.3.4 WIRING RULES

- Attach the antenna before powering the node. Transmitting without an antenna can damage the SX1262.
- The pigtail is RP-SMA female, so the antenna must be RP-SMA male. A standard SMA antenna will not connect properly. **VERIFY** the antenna's connector.
- Check battery polarity against the base board's markings before plugging in. JST-PH battery leads from different vendors are wired both ways.
- Fit a flyback diode across every motor, pump or solenoid.
- Tie the 5 V boost ground to the base board ground.

3.4 Sensor placement

Depths assume a bed about 30 cm deep. Scale them for other bins.

```

lid  [ pod: SCD41 . BME688 . LIS3DH ]  [ 40 mm fan -> vent ]
+-----+
|                                     |
|             headspace, about 8 to 10 cm             |
| ..... fresh feed layer ..... P5 | inlet holes |
| P1 (5 cm, centre)                    | with mesh  |
| P4 (15 cm, 3 cm from                |
|   sun-facing wall)                   |
| P2 (15 cm, centre)                   |
| P3 (25 cm, centre)                   |
+-----+
|             drainage tray (vermiwash)             |
+=====+
|             weighing platform: two boards, four load cells             |
+-----+

```

SENSOR	POSITION	WHAT IT CATCHES
P1	5 cm deep, centre	Heating of the fresh feed layer
P2	15 cm deep, centre	Main worm zone
P3	25 cm deep, centre	Heat and waterlogging near the bottom
P4	15 cm deep, 3 cm from the sun-facing wall	Heat through the bin wall
P5	In the latest feeding zone (moved at each feeding)	Heating of newly added material
M	Moisture sensor, 15 cm deep	Bed moisture
TMP117	Shaded, ventilated spot about 1 m above ground, away from the bin wall	Ambient reference

3.5 Lid pod and respiration hardware

- A small IP65 box fixed under the lid. Its side openings are covered with ePTFE membrane, which lets air and gas through but not liquid water.
- The SCD41 and BME688 face the headspace through the membrane. The LIS3DH is fixed rigidly to the lid.
- A 40 mm fan in a lid vent pushes headspace air out during the flush phase. Inlet holes (about four holes of 6 mm) are covered with fine mesh to keep flies out.
- During measurement the fan is off. The small inlet holes keep leakage low and similar from cycle to cycle, so the respiration trend is reliable even though the absolute value is approximate.
- The SCD41 is rated for non-condensing conditions. The membrane and flush reduce condensation. The firmware flags the pod if relative humidity stays above 95% for more than 12 hours.

3.6 Enclosures and cable entries

A PG7 gland seals one round cable of 3 to 6.5 mm.

GLAND	BOX	CABLE
G1 to G5	Node box	DS18B20 probes, one each
G6	Node box	Lid pod harness (I2C, fan, interrupt)
G7	Node box	Load cell cable
G8	Node box	Moisture sensor
G9	Node box	Solar panel
G10	Node box	Pump control (P1)
G11	Lid pod	Harness from the node box

Eleven glands are needed: four owned, eight more ordered (one spare).

- Seal the capacitive moisture sensor's cut board edges and circuitry with epoxy or nail polish, then heat-shrink over the electronics.
- Keep the node box outside the bin, on its shaded side, with a desiccant sachet inside.
- Mount the antenna vertically, above the bin rim.

3.7 Gateway hardware

- CM4 on a carrier board with the heatsink and a 5 V 3 A supply.
- Connect the LoRa HAT by USB through its on-board CP2102 at first (appears as `/dev/ttyUSB0`). This works on any Linux machine, which keeps the laptop fallback open. It can move to the 40-pin header (`/dev/serial0`) later.
- The HAT's M0 and M1 jumpers select configuration or transmission mode. Follow the Waveshare wiki.
- Ethernet (CAT6) for internet and maintenance. The CM4's WiFi runs as a hotspot for phones.
- Buzzer and LED on GPIO pins for local alerts.
- Timekeeping from the carrier's RTC or a DS3231, corrected by NTP when online.
- Keep the gateway indoors, out of direct sun, and log the processor temperature.
- Power from mains with a power bank or small UPS. Solar power for the gateway is out of scope for R0.

3.8 Power budget

Energy in. Charge current is limited by the panel to about 100 mA at most. With 4 to 5 useful sun hours outside the monsoon, and losses from angle, heat and dust, plan on roughly 250 to 350 mAh on a sunny day. Monsoon days deliver much less.

Energy out (estimates on the battery side; confirm in test T-PWR-01).

LOAD	DUTY	EST. MAH PER DAY
Node asleep, sensors off (about 0.05 mA)	Continuous	1.2
Sampling (sensors on about 3 s at about 10 mA)	288 per day	2.4
LoRa transmit (about 110 mA for about 0.29 s at SF9)	288 per day	2.6
BME688 gas measurement, ultra-low-power mode	Every 5 min	2
Weight unit, power-gated (about 10 mA for 2 s)	288 per day	1.6
SCD41 measurement phase (about 15 mA for 10 min)	6 per day	15
Fan flush (about 150 mA at 5 V for 2 min, through the boost converter)	6 per day	45
Boost converter idle	Continuous	3
Total		about 73

Autonomy. A 2,600 mAh cell with about 2,000 mAh usable gives roughly 25 days without sun at this load, well above the 7 days in NFR-01. Hot-mode sampling adds under 10 mAh per day. The pump is excluded and needs its own supply.

Low-battery policy.

BATTERY	BEHAVIOUR
3.6 V or more	Normal
3.4 to 3.6 V	Respiration cycle every 8 h
3.3 to 3.4 V	Sampling every 15 min; respiration and actuators off; low-battery flag set
Below 3.3 V	Hourly heartbeat only

4. Radio link

4.1 Mode and parameters

The MVP uses LoRa point-to-point (P2P), not LoRaWAN. There is one gateway, no network server is needed, and a single-channel HAT cannot act as a LoRaWAN gateway. LoRaWAN remains an option later with a multi-channel gateway.

PARAMETER	INITIAL VALUE	NOTES
Frequency	865.0625 MHz	Within the license-exempt 865 to 867 MHz band; confirm current power limits
Spreading factor	SF9	SF10 to SF12 for longer links
Bandwidth	125 kHz	
Coding rate	4/5	
Preamble	8 symbols	
Sync word	0x12 (private)	Must match the HAT (section 4.2)
Transmit power	14 dBm to start	Raise only if the range test needs it and rules allow
Periodic payload	44 bytes	About 0.29 s on air at SF9 (Appendix A)

4.2 HAT compatibility (risk R2)

The WaveShare SX1262 868M LoRa HAT is a UART "wireless serial port" module. It runs its own firmware, is configured through M0 and M1 jumpers and register commands, and may add its own packet framing. The RAK4631 drives its SX1262 directly. Interoperability is therefore not guaranteed and must be proven first.

Spike test (days 1 to 3)

1. Configure the HAT, following the WaveShare wiki, for transparent mode on a fixed channel in the 865 to 867 MHz range, lowest air data rate, and no encryption key.
2. On the RAK4631, run a RadioLib receiver that sweeps SF7 to SF12, bandwidths 125 and 250 kHz, and sync words 0x12 and 0x34, printing every packet received. Transmit repeatedly from the HAT.
3. Once packets arrive, lock the parameters, test the reverse direction, and document any framing bytes the HAT adds.
4. If no reliable two-way link exists by day 3, switch to a fallback.

Fallbacks, in order

- **F1.** A second RAK4631 on a WisBlock Base, connected by USB, as the gateway's radio modem. The same radio runs at both ends, so compatibility is guaranteed. About ₹4,600.
- **F2.** An SPI SX1262 module driven directly from the gateway.

The gateway's radio service hides the radio behind one interface, so nothing else in the software changes.

4.3 Uplink payloads

All multi-byte fields are little-endian.

Type 0x01, periodic reading (44 bytes)

OFFSET	SIZE	FIELD	TYPE	SCALE OR MEANING
0	1	version_type	uint8	High nibble protocol version (1); low nibble type (1)
1	2	node_id	uint16	
3	2	seq	uint16	Increments every packet
5	1	flags	uint8	bit 0 hot mode, 1 low battery, 2 pod humidity high, 3 lid open, 4 misting, 5 buffered record
6	2	age_s	uint16	Seconds since the sample was taken
8	2	batt_mv	uint16	Millivolts
10	10	t_probe[5]	int16 x 5	0.01 °C; 0x8000 means fault
20	2	t_ambient	int16	0.01 °C
22	2	rh	uint16	0.01 %
24	2	pressure	uint16	0.1 hPa
26	4	gas_res	uint32	Ohms
30	2	moisture_raw	uint16	Millivolts at the ADC
32	4	mass	int32	Grams
36	2	co2_last	uint16	ppm, last valid reading
38	1	fault_bits	uint8	bits 0 to 4 probe faults, 5 I2C fault, 6 weight fault, 7 moisture fault
39	1	rss_i_last_ack	int8	dBm of the last acknowledgement received
40	4	tag	bytes	First 4 bytes of HMAC-SHA256(node key, bytes 0 to 39)

Type 0x02, respiration result (28 bytes)

OFFSET	SIZE	FIELD	TYPE	SCALE OR MEANING
0	6	header		version_type, node_id, seq, flags as above
6	2	age_s	uint16	Seconds since the measurement ended
8	2	slope	int16	0.1 ppm per minute
10	2	r2	uint16	0.001
12	1	n_points	uint8	Points used in the fit
13	2	co2_start	uint16	ppm
15	2	co2_end	uint16	ppm
17	2	t_bed_mean	int16	0.01 °C, mean of the five probes
19	4	mass	int32	Grams

OFFSET	SIZE	FIELD	TYPE	SCALE OR MEANING
23	1	resp_flags	uint8	bit 0 saturated, 1 low R ² , 2 aborted by lid opening
24	4	tag	bytes	HMAC tag over bytes 0 to 23

Type 0x03, event (21 bytes)

OFFSET	SIZE	FIELD	TYPE	SCALE OR MEANING
0	6	header		As above
6	2	age_s	uint16	Seconds since the event
8	1	event_code	uint8	1 lid open, 2 lid closed with mass change, 5 unexplained mass change, 6 mist start, 7 mist stop, 8 failsafe triggered, 9 reboot
9	4	mass_before	int32	Grams
13	4	mass_after	int32	Grams
17	4	tag	bytes	HMAC tag over bytes 0 to 16

Type 0x08, downlink acknowledgement and command (19 bytes)

OFFSET	SIZE	FIELD	TYPE	SCALE OR MEANING
0	1	version_type	uint8	Type 8
1	2	node_id	uint16	
3	2	ack_seq	uint16	Sequence number being acknowledged
5	4	epoch	uint32	Gateway UTC time, seconds
9	1	cmd	uint8	0 none, 1 mist, 2 start respiration cycle, 3 set configuration value, 4 reboot
10	2	arg	uint16	For mist: seconds
12	1	cfg_key	uint8	Configuration parameter ID
13	2	cfg_value	int16	New value
15	4	tag	bytes	HMAC tag over bytes 0 to 14

4.4 Security

- Each node has a 32-byte HMAC key, set at flashing and stored in the gateway configuration. Keys are never committed to the repository.
- The 4-byte tag stops casual spoofing and corruption. Payloads are not encrypted because the data is not sensitive.
- Replay protection: the gateway rejects a sequence number already seen for that node within a window of 256. The node saves its sequence number to flash every 50 packets and adds 50 on boot, so it never repeats after a reset.
- Downlink commands carry a tag too, so a node ignores commands not signed with its key.

4.5 Reliability

- The gateway acknowledges every uplink. The node retries up to 2 times with a random 1 to 3 s delay, then stores the record (P1).
- Store-and-forward (P1): a ring buffer of 2,000 records, about 7 days at 5-minute sampling, in internal flash. After a successful acknowledgement the node sends up to 5 buffered records per

wake.

- The gateway logs signal strength per packet, and the health view warns when a node's link margin is low.

5. Node firmware

5.1 Toolchain and libraries

ITEM	CHOICE
Build	PlatformIO with the RAKwireless nRF52 platform, Arduino framework
Radio	RadioLib, or the SX126x-Arduino library, in P2P mode
DS18B20	OneWire and DallasTemperature
TMP117	SparkFun TMP117 library
BME688	Bosch BSEC2 in ultra-low-power mode, or the BME68x driver
SCD41	Sensirion I2C SCD4x library
LIS3DH	Adafruit LIS3DH library
Weight unit	M5Stack Weight I2C library or direct register access
Signing	Small software HMAC-SHA256 implementation
Storage	LittleFS on internal flash: configuration, sequence number, buffer

5.2 States

STATE	ENTERED ON	WORK	NEXT
BOOT	Power-on or watchdog reset	Load configuration and sequence number; sensor self-test; send reboot event	SLEEP
SLEEP	End of any task	Radio asleep; 3V3_S off; timer and LIS3DH interrupt armed	Timer or interrupt
SAMPLE	Sample timer	3V3_S on; read all sensors (median of 3 for moisture and mass); 3V3_S off	EVALUATE
EVALUATE	After SAMPLE	Edge failsafe rules (5.4); hot mode check; low-battery policy	TX
TX	Packet ready	Transmit, then listen 2 s for acknowledgement	APPLY, or retry
APPLY	Command received	Set time; run command; update configuration	SLEEP
RESP_FLUSH	Respiration timer	Fan on for 120 s	RESP_MEASURE
RESP_MEASURE	Flush done	Fan off; SCD41 in periodic mode for 600 s; fit slope	TX (type 0x02)
LID_OPEN	LIS3DH interrupt, lid tilted more than 30°	Record mass before (median of last 3 readings); abort any respiration run	LID_CLOSED
LID_CLOSED	Lid back within 5° of its closed position for 5 s	Wait 30 s to settle; read mass after (median of 5)	TX (type 0x03)
MIST (P1)	Failsafe rule or command	Pump on for the set time; start lockout timer	SLEEP

5.3 Schedules

TASK	DEFAULT	ALLOWED RANGE
Sampling	300 s	60 to 1,800 s
Hot mode sampling (any probe at 30°C or more)	60 s	30 to 300 s
Respiration cycle	Every 4 h	1 to 12 h
Flush	120 s	30 to 300 s
Measure	600 s	180 to 900 s
BME688 gas reading	Every 300 s	Fixed by BSEC mode
Heartbeat at critical battery	3,600 s	Fixed

5.4 Edge failsafe rules

These run on the node whether or not the gateway is reachable. Values are initial and set by configuration.

RULE	CONDITION	ACTION
EF-1 Heat, bed not wet	Any probe at 33°C or more, and moisture below 75%	Mist 20 s (P1); lockout 30 min; event 8
EF-2 Heat, bed wet	Any probe at 33°C or more, and moisture 75% or more	Hot mode and flag only; misting a wet bed makes it worse
EF-3 Dry	Moisture below 55% for 2 samples in a row	Mist 15 s (P1); lockout 60 min
EF-4 Pump safety	More than 3 min of misting in 1 h	Disable the pump until the gateway re-enables it

5.5 Robustness

- Hardware watchdog at 30 s.
- Every sensor read has a timeout. A failing sensor sets its fault bit and never blocks the loop.
- A DS18B20 reading of 85.00°C right after power-up, or -127°C (disconnected), is treated as a fault.
- Configuration and sequence number persist across resets.
- Firmware updates over Bluetooth without opening the box (P2).

6. Gateway software

6.1 Stack

LAYER	CHOICE
Operating system	Raspberry Pi OS Lite, 64-bit, on eMMC
Language	Python 3.11
Messaging	Mosquitto MQTT, bound to localhost
Storage	SQLite in WAL mode
API	FastAPI with Uvicorn; WebSocket for live updates

LAYER	CHOICE
Analysis	NumPy and SciPy
Dashboard	Installable web app (PWA), uPlot charts, translation files for Gujarati, Hindi and English
GPIO	gpiozero for buzzer and LED
Outbound alerts	Telegram Bot API over HTTPS when online
Processes	systemd services, journald logs
Networking	NetworkManager hotspot on wlan0; Ethernet uplink on eth0; mDNS name vermikendra.local

6.2 Services

SERVICE	RESPONSIBILITY
vk-radio	Owns the LoRa radio; checks tags and sequence numbers; decodes; publishes to <code>vk/{site}/{node}/up</code> ; sends acknowledgements and queued commands from <code>vk/{site}/{node}/down</code>
vk-ingest	Stores readings, events and respiration runs
vk-engine	Calibration, event classification, heat forecast, readiness, alert rules; publishes to <code>vk/{site}/alerts</code>
vk-api	REST and WebSocket API; serves the dashboard
vk-notify	Buzzer, LED, and a Telegram outbox with retry
vk-replay	Replays a recorded dataset into MQTT at 1x to 600x speed (FR-12)
vk-sync (P2)	Sends queued records to a cloud endpoint when online

6.3 Decoded message

Example of a decoded reading as published to MQTT. Values are illustrative.

```
{
  "site": "rru-campus",
  "node": 1,
  "bin": "bin-01",
  "type": "reading",
  "seq": 4182,
  "ts": "2026-10-03T14:05:12+05:30",
  "battery_mv": 3912,
  "probes_c": [27.41, 29.86, 26.10, 31.02, 33.40],
  "ambient_c": 34.18,
  "rh_pct": 91.2,
  "pressure_hpa": 1006.4,
  "gas_ohm": 48210,
  "moisture_pct": 71.5,
  "mass_kg": 42.816,
  "co2_ppm": 1840,
  "flags": ["hot_mode"],
  "faults": [],
  "rssi_dbm": -97,
  "snr_db": 6.5
}
```

6.4 Data model

TABLE	KEY COLUMNS
sites	id, name, village, district, latitude, longitude

TABLE	KEY COLUMNS
bins	id, site_id, name, volume_l, bed_depth_cm, headspace_l, tare_kg, status
nodes	id, bin_id, key_id, firmware_version, last_seen, battery_mv
calibrations	node_id, sensor, parameters (JSON), valid_from
readings	node_id, ts, seq, probe_1 to probe_5, ambient, rh, pressure, gas_ohm, moisture_raw, moisture_pct, mass_g, co2_ppm, battery_mv, rssi, snr, flags, faults
respiration_runs	id, bin_id, ts_start, slope_ppm_min, r2, n_points, index_value, bed_temp_c, valid
events	id, bin_id, ts, type, mass_before_g, mass_after_g, delta_g, source (auto or manual), note
alerts	id, bin_id, ts_open, ts_clear, rule_id, severity, value, message_key, ack_user, ack_ts
batches	id, bin_id, ts_start, ts_harvest, input_g, output_g, readiness_history (JSON)
users	id, name, role (operator, supervisor, admin), phone, language, pin_hash
commands	id, node_id, ts, type, argument, status

Retention: full-resolution readings for 90 days, then 1-hour averages. One bin produces roughly 60 KB of readings per day, so storage is not a constraint on 32 GB.

6.5 REST API (version 1)

METHOD	PATH	PURPOSE
GET	/api/v1/bins	All bins with status, top alert and readiness
GET	/api/v1/bins/{id}	One bin with its latest reading
GET	/api/v1/bins/{id}/readings?from=&to=&step=	Time series, downsampled to the requested step
GET	/api/v1/bins/{id}/respiration?from=&to=	Respiration runs
GET	/api/v1/bins/{id}/events	Events
POST	/api/v1/bins/{id}/events	Manual event (FR-14)
GET	/api/v1/bins/{id}/readiness	Status, conditions met, days-to-ready estimate (P1)
GET	/api/v1/alerts?state=open	Open alerts
POST	/api/v1/alerts/{id}/ack	Acknowledge an alert (PIN)
POST	/api/v1/nodes/{id}/commands	Mist, start respiration cycle, reboot (PIN)
GET, PUT	/api/v1/bins/{id}/config	Thresholds and schedules (PIN for changes)
GET	/api/v1/impact?from=&to=	Waste in, compost out, batches
GET	/api/v1/export.csv?bin=&from=&to=	CSV export (FR-10)
GET	/api/v1/health	Gateway and node health (FR-13)
WS	/ws/live	Readings, alerts and events as they arrive

6.6 Dashboard

SCREEN	CONTENT
Overview	One card per bin: status (colour, icon and word), last update, top alert, readiness; site-wide banner for critical alerts

SCREEN	CONTENT
Bin detail	Status and the one action to take; probe temperatures drawn at their depths; charts for temperature, moisture, respiration and mass; event timeline
Alerts	Open and recent alerts with actions; acknowledge
Records	Batches, inputs, outputs, totals; CSV export
Health	Battery, signal, last seen, sensor faults; gateway disk and temperature
Settings	Language, bins, thresholds, users, Telegram (PIN protected)

- Status words are fixed and translated: **Good, Needs attention, Act now, Offline.**
- Action text is stored as translation keys, for example `action.heat.shade` and `action.dry.water`.
- Must work on a 5-inch Android phone in Chrome, load within 3 s over the hotspot, and keep the last data viewable when the phone is out of range.

6.7 Networking

- Hotspot name `Vermikendra-{site}`, WPA2. The gateway is at 10.42.0.1 (NetworkManager's shared-mode default), so the dashboard is at <http://10.42.0.1> or <http://vermikendra.local>.
- The Ethernet uplink is optional. Services listen only on the hotspot and localhost; no inbound ports are opened to the internet.
- Remote maintenance (P2) uses an outbound tunnel only.

6.8 Time

The gateway timestamps each record as its receive time minus `age_s`. Gateway time comes from the RTC and is corrected by NTP when online. Acknowledgements carry the gateway time so nodes can stamp buffered records.

7. Algorithms

All thresholds are initial values, stored in configuration, and will be tuned with pilot data.

7.1 Thresholds

PARAMETER	GOOD	WARNING	CRITICAL
Bed temperature, any probe	12 to 30°C	30 to 33°C, or 5 to 12°C	33°C or more, or below 5°C
Moisture content	60 to 80%	55 to 60%, or 80 to 85%	Below 55%, or above 85%
Foul air ratio (gas resistance divided by its 24 h median) while RH is above 90%	0.8 or more	0.6 to 0.8	Below 0.6 for 1 h
Battery	3.6 V or more	3.4 to 3.6 V	Below 3.4 V
Time since last packet	Under 15 min	15 to 30 min	Over 30 min

Worm losses become likely around 35°C, so the critical line sits at 33°C to leave time to act.

7.2 Probe calibration

Each probe i gets an offset from the three-point water bath in section 8.1. If residuals exceed 0.2°C , a gain is fitted as well.

```
T_cal[i] = gain[i] * T_raw[i] + offset[i]
```

7.3 Heat-risk forecast

For each probe, fit a least-squares line to the last 30 minutes of readings (at least 5 points), giving the current temperature T and slope s in $^{\circ}\text{C}$ per hour.

```
if T >= T_crit:                critical (HEAT_NOW)
elif s > 0.3:
    t_cross = (T_crit - T) / s  # hours until the probe reaches T_crit
    if t_cross <= 2:            critical (HEAT_FORECAST)
    elif t_cross <= 6:         warning (HEAT_FORECAST)
```

The alert clears when T is at least 1.5°C below T_{crit} and s is 0 or less for 30 minutes.

7.4 Moisture

- Two-point calibration per sensor in the real bedding: `raw_dry` (air-dried bedding) and `raw_wet` (saturated, drained for 10 minutes). Capacitive sensor output falls as moisture rises.
- Wetness index: $w = (\text{raw_dry} - \text{raw}) / (\text{raw_dry} - \text{raw_wet})$, from 0 (dry) to 1 (saturated).
- Map w to gravimetric moisture content with a linear fit from 4 to 6 bedding samples, each weighed wet, dried to constant weight, and weighed again. Moisture content (wet basis) = $(\text{wet mass} - \text{dry mass}) / \text{wet mass} \times 100$.

The thresholds in 7.1 refer to this moisture content.

7.5 Respiration rate

For each cycle, using SCD41 readings every 5 s during the measure phase:

- Discard the first 60 s, while air settles after the fan stops, and any reading of 4,800 ppm or more.
- Fit a straight line $c(t) = a + s * t$ to the remaining points (up to 480 s). The slope s is in ppm per minute.
- The run is valid if it has at least 20 points and R^2 of at least 0.90. Otherwise it is flagged low quality.
- Convert to a mass-normalised index, in mg of CO_2 -carbon per kg of bed per hour:

```
n_air   = P * (V_head / 1000) / (8.314 * T_K)      # mol of air in headspace
R_index = s * 1e-6 * n_air * 12.01 * 1000 * 60 / m_bed

P       headspace pressure, Pa (from BME688)
V_head  headspace volume, litres (entered at setup, updated at harvest)
T_K     headspace temperature, kelvin
m_bed   bed mass, kg = platform mass - bin tare
```

- Optional normalisation to 25°C , assuming respiration doubles every 10°C : $R_{25} = R_{\text{index}} * 2^{((25 - T_{\text{bed}}) / 10)}$.

Because of leakage the index is lower than the true rate. It is used only as a trend within the same bin.

7.6 Feeding and harvest events

The node reports mass before and after each lid opening. The gateway classifies the change.

CONDITION	CLASSIFICATION
Change of +0.3 kg or more	Input (feeding)
Change of -0.3 kg or less	Output (harvest or removal)
Change smaller than 0.3 kg either way	Inspection (lid opened, nothing added)
Step of 0.5 kg or more between readings with no lid event	Unexplained change: shown for review, not counted

- Slow mass loss between events (evaporation and decomposition) is tracked as a daily loss rate. A rising loss rate together with falling moisture strengthens the dry alert.
- At each harvest the operator is prompted to confirm the empty-platform reading, which resets the tare (P1).

7.7 Foul air indicator (P1)

- Baseline B is the rolling 24-hour median of BME688 gas resistance.
- Ratio r is the current gas resistance divided by B.
- Raise the indicator when r is below 0.6 and RH is above 90% for at least 1 hour, excluding the 2 hours after a feeding (fresh feed briefly releases gases).
- Action: "Turn the bed gently and add dry bedding. Check the drainage holes."

The BME688 is a metal-oxide sensor that responds to many gases. This is a risk indicator, not a gas concentration measurement.

7.8 Readiness

Version 1, rules (P0). The status is **Ready** when all four conditions hold:

1. At least 30 days since the batch started and at least 10 days since the last input.
2. All probes within $\pm 2^{\circ}\text{C}$ of ambient for 3 consecutive days, meaning the bed is no longer heating itself.
3. The respiration index has fallen at least 70% below its peak for the batch, and its 5-day trend is flat.
4. Moisture has stayed in the good range for the last 3 days.

The status is **Maturing** when conditions 1 and 2 hold but 3 does not, and **Not ready** otherwise. The dashboard lists which conditions are met.

Version 2, prediction (P1). Fit an exponential decay to valid respiration runs since the last input, requiring at least 12 runs spanning 48 hours or more:

$$R(t) = R_{\text{inf}} + (R_0 - R_{\text{inf}}) * \exp(-k * t) \quad \# k > 0, R_{\text{inf}} \geq 0$$

t_{ready} : the t at which $R(t) = R_{\text{inf}} + 0.1 * (R_0 - R_{\text{inf}})$

Report days-to-ready with a range from the fit's parameter uncertainty, using a bootstrap if the fit is unstable. Hide the estimate while the range is wider than ± 7 days. Once several batches exist (R2), train a gradient-boosted model on feedstock type, temperature history, moisture and respiration curve shape, validated by leaving one batch out at a time.

7.9 Alert rules

RULE	CONDITION	SEVERITY	ACTION SHOWN	CLEARs WHEN
HEAT_NOW	Any probe at 33°C or more	Critical	Shade the bin; do not add fresh dung; mist if the bed is not wet	Probe at or below 31.5°C for 30 min
HEAT_FORECAST	Forecast crossing within 2 h (critical) or 6 h (warning)	Critical Warning	As above, earlier. If auto-misting is on and moisture is below 75%, the gateway sends a mist command	Forecast no longer met for 30 min
HEAT_WARM	Any probe between 30 and 33°C for 30 min	Warning	Check shade and airflow	Probe at or below 29°C
COLD	Any probe below 12°C for 6 h (below 5°C is critical)	Warning Critical	Cover the bin at night; add bedding	Probe at 14°C or more
DRY	Moisture below 60% for 1 h (below 55% is critical)	Warning Critical	Sprinkle water evenly; check again in 1 h	Moisture at 62% or more
WET	Moisture above 80% for 6 h (above 85% is critical)	Warning Critical	Add dry bedding; check the drainage holes	Moisture at or below 78%
FOUL_AIR (P1)	Section 7.7	Warning	Turn the bed gently; add dry bedding	Ratio at 0.8 or more for 2 h
READY	Section 7.8	Info	Harvest when convenient	Status changes
NODE_OFFLINE	No packet for 30 min	Warning	Check node power and antenna	Packet received
BATTERY_LOW	Below 3.4 V	Warning	Clean the solar panel; check it gets sun	3.6 V or more
SENSOR_FAULT	A fault bit set for 3 samples in a row	Warning	Check the sensor's connection	3 good samples
POD_WET	Pod RH above 95% for 12 h	Info	Check the lid pod membrane	RH below 90%
RESP_SATURATED	3 saturated runs in a row	Info	Shorten the measure phase	A valid run

The same rule re-notifies for the same bin at most every 2 hours, unless its severity rises.

8. Calibration procedures

8.1 Temperature probes

1. Tie the five probe tips together with the TMP117, sealed in a thin plastic bag.
2. Prepare three stirred water baths at about 15°C, 25°C and 40°C. Let each settle for 5 minutes.
3. Record 10 readings in each bath and compute each probe's offset, plus gain if needed.
4. Accept when residuals are 0.2°C or less. Store the values in the calibration table.

8.2 Load cells

1. Assemble the platform on a flat, rigid base and level it.
2. Tare it empty, then add known masses of 1, 5, 10 and 20 kg (sealed water bottles checked on a kitchen scale).
3. Fit a linear gain. Accept a residual of 100 g or less at 20 kg.
4. Leave 5 kg on the platform for 12 hours and record creep and temperature drift.

8.3 Moisture sensor

Follow section 7.4 in the real bedding. Dry samples in the sun for 2 to 3 days, or in an oven at about 105°C, until their weight stops changing.

8.4 SCD41

1. Turn off automatic self-calibration. It assumes the sensor sees fresh air (about 400 ppm) at least once a week, which never happens inside a compost bin, so it would drift the baseline.
2. Run forced recalibration outdoors in open air, away from people and exhaust. Operate the sensor in its normal measurement mode for at least 3 minutes first, then set the reference to about 420 ppm.
3. Repeat every 2 weeks, and after any drop or firmware change. Log each recalibration.

8.5 BME688

Run the sensor for at least 48 hours before recording a baseline. Save the BSEC algorithm state to flash every 6 hours so the baseline survives resets.

9. Test plan

ID	LEVEL	PROCEDURE	PASS CRITERIA	COVERS
T-SEN-01	Bench	Probe calibration (8.1)	Within $\pm 0.5^{\circ}\text{C}$ of TMP117; target $\pm 0.2^{\circ}\text{C}$	FR-01
T-SEN-02	Bench	Unplug one probe; power-cycle the node	Fault flagged within 3 samples; other sensors keep reporting	FR-01, FR-13
T-SEN-03	Bench	Load cell calibration (8.2)	100 g or less residual at 20 kg	FR-05
T-SEN-04	Bench	Moisture calibration (8.3)	Within ± 10 points of gravimetric values	FR-03
T-SEN-05	Bench	Respiration cycle on fresh kitchen and dung mix vs mature compost, same container and mass	Fresh rate at least twice the mature rate; R^2 of 0.9 or more	FR-04
T-RAD-01	Bench	HAT spike test (4.2)	Two-way link, or fallback chosen, by day 3	Risk R2
T-RAD-02	Field	Range walk at 100 m, 500 m, 1 km, 2 km and 3 km; 100 packets each; antennas 1.5 m above ground	At least 95% of packets delivered at 1 km	NFR-02
T-PWR-01	Bench	Measure current in each state with a multimeter or USB power meter; recompute the budget	100 mAh per day or less	NFR-01
T-PWR-02	Field	Cover the panel and run on battery for 72 h	No brownout; projected autonomy of 7 days or more	NFR-01
T-ENV-01	Field	Splash the closed node box with water for 5 min	No water inside	NFR-06
T-ENV-02	Field	Node box in shade through the hottest afternoon	Box temperature logged; no resets	NFR-06, Risk R11
T-INT-01	System	Put one probe in 40°C water	Critical alert on dashboard within 2 min; buzzer sounds; Telegram message if online	FR-02, NFR-04
T-INT-02	System	Open lid, add 2.0 kg of weighed waste, close lid	Input event of 2.0 ± 0.2 kg within 90 s	FR-05
T-INT-03	System	Switch the gateway off for 12 h (24 h for P1)	P0: gap flagged. P1: full backfill	NFR-03, FR-19
T-INT-04	System	Disconnect the gateway from the internet	All P0 features still work	NFR-05, FR-07

ID	LEVEL	PROCEDURE	PASS CRITERIA	COVERS
T-INT-05	System	Send a packet with a wrong tag, and replay an old packet	Both rejected and logged	NFR-09
T-INT-06	System	Run replay mode on a laptop with no hardware	Dashboard shows the full dataset	FR-12
T-UX-01	Users	Three people outside the team use the Gujarati interface with no training	Each finds the bin needing attention, and its action, within 10 s	NFR-07, FR-15
T-PIL-01	Pilot	Two bins for 30 days: one monitored and acted on, one checked daily by hand thermometer	Complete records; comparison documented	PRD 9.1

10. Submission repository

The hackathon requires a public repository with complete source, a README with setup steps, and architecture documentation. Planned layout:

```

vermikendra/
  README.md      setup, hardware, replay mode, architecture overview
  LICENSE
  docs/          PRD, TRD, calibration notes, wiring diagrams, photos
  firmware/node/ PlatformIO project for the RAK4631
  gateway/       vk-radio, vk-ingest, vk-engine, vk-api, vk-notify, vk-replay
  gateway/dashboard/ web app and translation files (gu, hi, en)
  gateway/config/ example configuration, no secrets
  hardware/      bill of materials, pin map, enclosure drilling templates
  data/sample/   recorded datasets for replay mode
  tools/         packet simulator, airtime calculator

```

- Keep the main branch demo-ready. Work on branches; one teammate reviews each merge.
- Automated checks on every push: gateway tests, firmware build, lint.
- Never commit HMAC keys or the Telegram token.
- Tag releases R0, R1 and R2 to match the PRD release plan.
- Decide on a provisional patent before the repository becomes public (Risk R10).

11. Risk register

ID	RISK	LIKELIHOOD	IMPACT	MITIGATION	OWNER, CHECK BY
R1	No CM4 carrier board	High	Blocks the gateway	Borrow from a lab or order now; run the gateway software on a laptop with the HAT over USB in the meantime	Hardware, 16 Sep
R2	HAT and RAK4631 cannot talk	Medium	Blocks the radio link	Spike test (4.2); fallback F1 or F2	Hardware, 17 Sep
R3	Condensation or ammonia damages pod sensors	Medium	Data loss	ePTFE membrane, flush fan, humidity monitoring, spare BME688	Hardware
R4	SCD41 saturates or drifts	Medium	Poor respiration data	Short measure window; discard readings at 4,800 ppm or more; self-calibration off; recalibrate every 2 weeks	Firmware
R5	Worms die early	Medium	No dataset	Aged feedstock, shade, two bins, spare worms; a failure the system detects is still evidence	Field

ID	RISK	LIKELIHOOD	IMPACT	MITIGATION	OWNER, CHECK BY
R6	Too little data for prediction by 14 Oct	High	Weaker AI claim	Rules-based readiness in R0; prediction in R1; bench respiration test for the video	Gateway
R7	Solar panel too small for actuators	High	No automatic misting	Actuators on a separate supply; the small panel powers the node only	Hardware
R8	Load cell drift outdoors	Medium	Wrong event masses	Short-window mass differences for events; creep test; tare at harvest	Firmware
R9	Eligibility question	Low to medium	Disqualification	Email the MY Bharat helpdesk this week	Team lead, 18 Sep
R10	Public repository counts as disclosure	Medium	A patent could be blocked	Decide on a provisional filing before the repository goes public	Team lead, 30 Sep
R11	Weight unit above its rated 40°C	Medium	Resets or drift in summer	Shaded, ventilated node box; log box temperature	Hardware
R12	Schedule slip	Medium	Incomplete submission	Strict P0 scope; demo to each other weekly; submit a day early	All

12. Build plan

Three workstreams, one owner each; everyone joins integration tests.

- **A. Hardware and firmware**
- **B. Gateway, data and algorithms**
- **C. Product, field pilot, dashboard text, deck and video**

WEEK	DATES	A: HARDWARE AND FIRMWARE	B: GATEWAY AND DATA	C: PRODUCT AND FIELD
W0	14 to 20 Sep	Identify base board and HAT; order parts (3.2); radio spike test	Gateway software on a laptop with the HAT over USB; packet decoder and simulator	Register on MY Bharat; take the quiz; eligibility email; source worms, bins and aged dung
W1	21 to 27 Sep	All sensors reading on USB power; probe and load cell calibration	Ingest, database, API skeleton; replay mode	Set up both bins; USB-powered node logging on bin 1 by 27 Sep; manual register for bin 2
W2	28 Sep to 4 Oct	Enclosures, glands, solar power, lid pod, respiration cycle	Alert rules, heat forecast, events, readiness version 1	Dashboard screens; translated status and action text; user test plan
W3	5 to 11 Oct	Range test, power measurements, 72 h battery test	Charts, CSV export, health view; Telegram if time allows	Usability test with 3 people; deck draft; video script and filming
W4	12 to 14 Oct	Freeze firmware for R0	Freeze gateway for R0; README and architecture documentation	Final deck and video; submit on 14 Oct
After	From 16 Oct	P1: misting, store-and-forward	P1: prediction version 2, foul air indicator, batches	State round rehearsal; field partner outreach

Appendix A: Time on air

Semtech time-on-air formula for a 44-byte payload, 125 kHz bandwidth, coding rate 4/5, explicit header, CRC on, 8-symbol preamble. Low data rate optimisation is on at SF12 only.

SPREADING FACTOR	SYMBOL TIME	TIME ON AIR
SF7	1.024 ms	about 0.09 s
SF9	4.096 ms	about 0.29 s
SF10	8.192 ms	about 0.53 s
SF12	32.768 ms	about 2.1 s

Appendix B: Datasheets and references

- Sensirion SCD4x datasheet: sensirion.com
- RAKwireless RAK19007 WisBlock Base datasheet: docs.rakwireless.com
- Waveshare SX1262 868M LoRa HAT wiki: waveshare.com
- M5Stack Weight I2C Unit documentation: docs.m5stack.com
- Hack for Social Cause 2027 rules: mybharat.gov.in